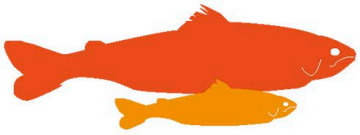
GENETICALLY MODIFIED ANIMALS

While genetically engineered crops are already being grown commercially in some parts of the world, most modified animals are still at the research stage. Genetically modified (GM) livestock are being bred to improve commercially important traits such as better growth rate, disease resistance, meat quality, or offspring survival rate. GM salmon, for example, have been bred to grow twice as fast as conventional salmon.

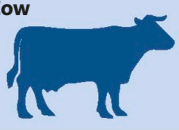


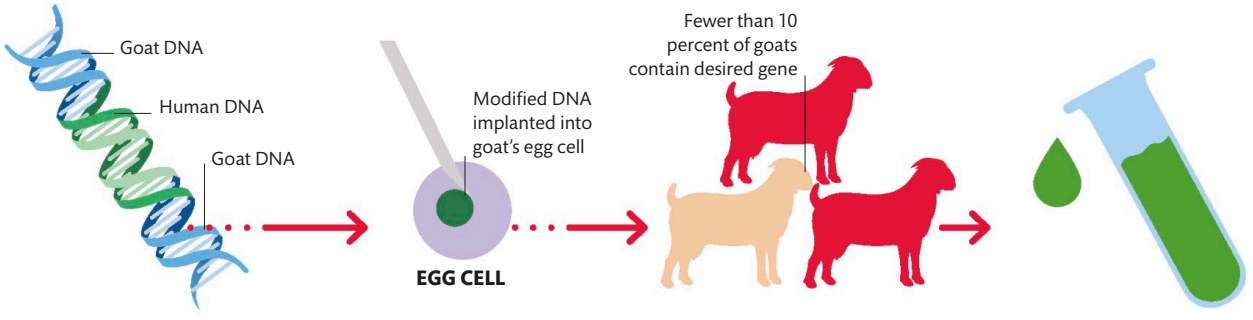
THE **FIRST GENETICALLY MODIFIED CROP** TO BE SOLD WAS A **TOMATO**



Transgenic animals

A few products are already produced by transgenic livestock, with others under development. Transgenic animals are animals that have had a gene from another species inserted into their DNA. One use of transgenic animals is to produce pharmaceutical goods. Raising animals is cheaper than setting up a pharmaceutical production line to make drugs, but developers are currently restricted to products that can be extracted from milk, eggs, or other products that do not harm the animal. The use of urine also has potential for investigation as it is not dependent on the sex or age of the animal.

ANIMAL	USES
Cow 	Transgenic cows can be used to create several products, such as milk containing human lactoferrin, a protein that can be used to treat infections. Scientists have also created genetically modified cows' milk that has a lower lactose content, making it suitable for people who are lactose intolerant.
Pig 	Scientists are researching how the genes of pigs could be edited so that the animals' organs could be made suitable for use in human organ transplants. Pigs have been genetically modified so that they produce phytase, an enzyme that reduces the pig's excretion of phosphorus, making its waste less polulting.
Goat 	Goats have been genetically modified to produce human antithrombin, a protein that prevents blood clotting (see below). Scientists have also created goats that are capable of producing silk in their milk by inserting the silk protein gene found in spiders into the goat's DNA.
Sheep 	Scientists have produced sheep that have high levels of omega-3 fatty acids in their meat by inserting a roundworm gene, linked to the production of the fatty acids, into the sheep genome. Sheep have also been genetically modified to carry the gene for Huntington's disease, allowing scientists to study the illness.



1 Modifying the DNA
A section of human DNA containing the code for the blood hormone antithrombin (which reduces clotting) is inserted into goat DNA.

2 Implanting the DNA
The modified strand of DNA is injected into the nucleus of a fertilized goat egg. It is then implanted into a female goat, who carries the embryo to term.

3 Testing the offspring
The offspring are tested to see whether they carry the antithrombin gene. Those that do are bred to form a herd of modified goats.

4 Extracting the protein
Milk from the modified goats is filtered and purified. In a year, a single goat can produce as much antithrombin as 90,000 blood donations.